



Semester II Examinations 2018/2019

Exam Code(s)	3BS9, 3BPT1, 4BS2, 4BME1, 1CSD1, 1CSD2, 1OA2
Exam(s)	Third Science, 3rd Bachelor of Science (Physics), Fourth Science, Fourth Mathematics and Education; MSc Data Analytics
Module(s)	Applied Statistics II
Module Code(s)	ST312
Paper No	1
Release to Library	Yes
External Examiner(s)	Prof. Simon Wilson
Internal Examiner(s)	Dr. Andrew Simpkin Prof. John Newell
Instructions	Answer FIVE questions
Duration	2 Hours
No. of Pages	13 pages
Discipline	Statistics
Requirements:	
Release in Exam Venue	Yes
MCQ	No
Statistical Tables	Relevant distribution tables are attached to this paper.
Other Materials Allowed	A calculator is allowed (non-programmable and not capable of storing text).

Question 1: [Total marks 15]

Hypericum which is extracted from the St. John's Wort plant has been known to help alleviate depressive symptoms.



In a recent study, 75 people presenting with mild to moderate depressive symptoms were recruited on a convenience basis from a local GP surgery. Hypericum extract was compared against a common antidepressant as well as a placebo. The Hamilton Depression Scale (HAMD) was used to assess depressive symptoms before the study began and again 6 weeks after study treatment began.

- a) Is this an observational or experimental study? Explain your answer.
(2 marks)
- b) Identify the response and explanatory variables, the treatments, and the experimental units.
(4 marks)
- c) Are the results generalisable from the sample of patients to the population of individuals with mild to moderate depressive symptoms? Why/why not?
(2 marks)
- d) How should the researchers assign the subjects to the three groups? Why? Explain the advantages of your proposed approach.
(3 marks)
- e) Explain how and why the experimenters might blind study participants to their treatment allocation
(4 marks)

Question 2: [Total marks 15]

Sports scientists for a Premier League soccer club are interested in reducing injury to their players. Two novel injury prevention methods – one being a novel functional warm-up, the second being training load control – are to be tested against standard training. The sports scientists trial these methods among their academy players, with three players from each of six teams chosen to participate in the study. The three players from each team were randomly assigned to the three treatments. After the training session knee flexibility (a continuous measure of injury susceptibility) was measured on a scale of 0 to 100%.

The table below shows knee flexibility, with the values given in a systematic order rather than the random order of the experiment.

	Team					
	1	2	3	4	5	6
Standard training	75	68	79	71	65	73
Functional warm up	86	81	92	79	78	90
Training Load Control	81	89	91	87	87	86

- a) List the factor of interest and any nuisance factors. (2 marks)
- b) What are the experimental units? (1 mark)
- c) Are there blocks in this design? If so, identify them. (2 marks)
- d) Why did the sports scientists conduct the experiment in this way? (2 marks)
- e) Has their approach been justified by the data? (2 marks)
- f) Describe how a randomization test could be used to see if the training regimens have any effect on flexibility, taking account of the possible between team variation. State the null and alternative hypotheses being tested and describe (briefly) how such a test would be implemented including how the associated p-value would be calculated. What would be the advantage of doing this rather than a standard ANOVA F-test? (No calculations are required.) (6 marks)

Question 3: [Total marks 35]

To investigate how the housing crisis is affecting rental costs in Ireland, researchers randomly sampled 10 semi-detached houses from each of four areas (Cork, Dublin, Galway, Mullingar) on two occasions (2007 and 2019). Monthly rental cost (in euro) was used to examine the effect of the housing crisis.

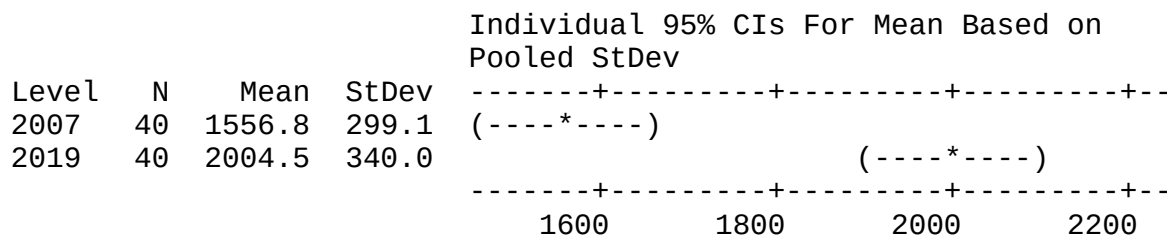
- a) An initial analysis using analysis of variance looked to see if there was any difference between 2007 and 2019 prices. Use the output below to test the hypothesis that there was no difference between the two measurement times. Clearly specify your null and alternative hypotheses, calculate an appropriate F-statistic and use the tables at the end of this paper to perform a test at the 5% level.

(4 marks)

One-way ANOVA: rent versus year

Source	DF	SS	MS	F	P
year	1	4009601	4009601	***	***
Error	78	7999067	102552		
Total	79	12008669			

S = 320.2 R-Sq = 33.39% R-Sq(adj) = 32.54%



Pooled StDev = 320.2

- b) An alternative would have been to use a two-sample t-test; what would have been the value of the t-test statistic?

(2 marks)

- c) Next they looked to see if there was any difference between the areas. Using the Minitab output on the next page, test the hypothesis that there are no differences between the rental areas. Again, clearly specify your null and alternative hypotheses, perform an F-test at the 5% level, and state your conclusions.

(4 marks)

One-way ANOVA: rent versus area

Source	DF	SS	MS	F	P
area	3	1763084	587695	4.36	0.007
Error	76	10245585	134810		
Total	79	12008669			

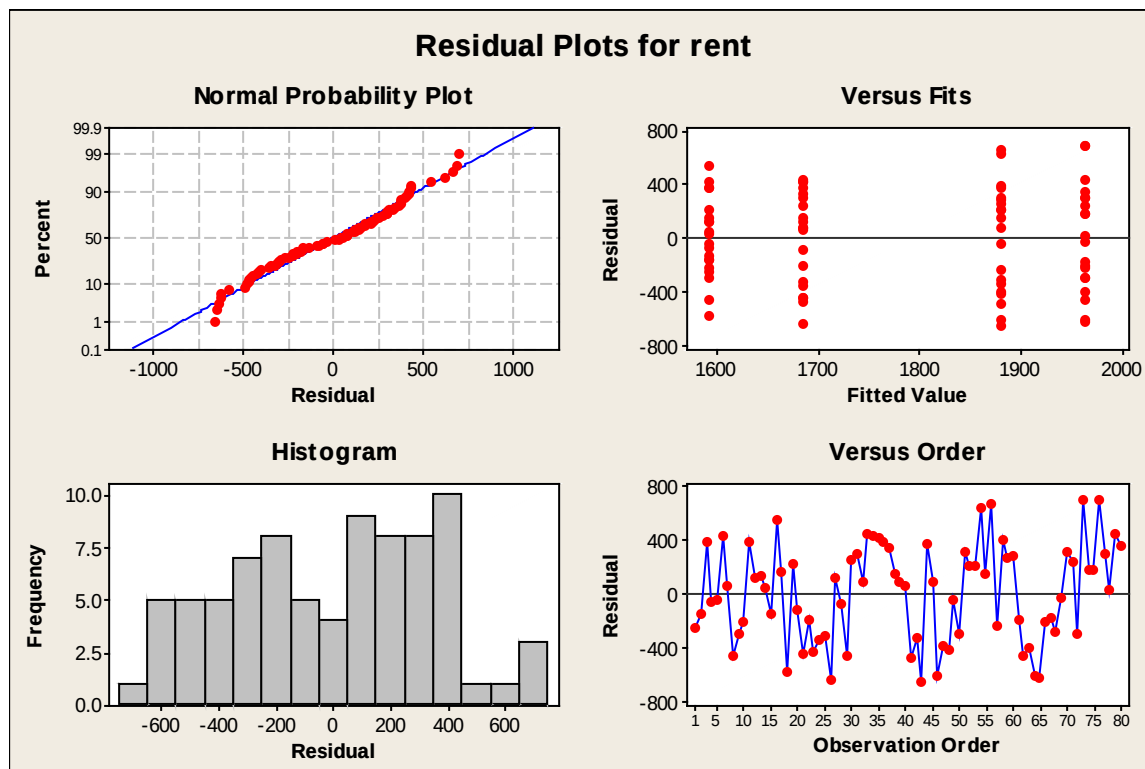
S = 367.2 R-Sq = 14.68% R-Sq(adj) = 11.31%

Individual 95% CIs For Mean Based on Pooled StDev			
Level	N	Mean	StDev
Cork	20	1685.0	349.3
Dublin	20	1963.5	400.5
Galway	20	1881.5	408.1
Mullingar	20	1592.5	300.5

Pooled StDev = 367.2

- d) Comment on the validity of the assumptions for this ANOVA based on the residual plots below. If you have reservations on the validity, suggest what you might do to improve matters.

(3 marks)



- e) Following on from the above ANOVA why does it make sense to perform a multiple comparisons procedure?
(1 mark)
- f) Use the following output to perform multiple comparisons. What are the familywise and individual type I error rates of your procedure? Group the different areas accordingly.
(3 marks)

Tukey Simultaneous Tests for Differences of Means

Difference of Levels	Difference of Means	SE of Difference	95% CI	T-Value	Adjusted P-Value
Dublin - Cork	279	116	(-27, 584)	2.40	0.086
Galway - Cork	197	116	(-109, 502)	1.69	0.335
Mullingar - Cork	-93	116	(-398, 213)	-0.80	0.856
Galway - Dublin	-82	116	(-387, 223)	-0.71	0.894
Mullingar - Dublin	-371	116	(-676, -66)	-3.20	0.011
Mullingar - Galway	-289	116	(-594, 16)	-2.49	0.070

Individual confidence level = 98.97%

- g) Cork, Dublin and Galway are cities, while Mullingar is not. Define a suitable contrast to compare the population mean of Mullingar with that of Cork, Dublin and Galway. Specify the null and alternative hypotheses being tested, obtain an estimate for the contrast statistic, calculate its standard error and the degrees of freedom for the test. Perform the test at the 5% significance level.
(6 marks)
- h) To consider both factors, year and area, a two-way ANOVA was performed with the output given below. What do you conclude? Relate your conclusions to your previous findings, in particular in part (a).
(4 marks)

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
area	3	1763084	587695	7.07	0.000
year	1	4009601	4009601	48.22	0.000
Error	75	6235984	83146		
Total	79	12008669			

- i) The model in the above output is just a main effects model. The tables below give the cell means (based on 10 observations per cell) and a partial table of interaction effects. Complete the entries in the interaction effects table.
(2 marks)

<i>Cell means</i>	<i>2007</i>	<i>2019</i>	<i>Row Mean</i>
<i>Cork</i>	1422	1948	1685
<i>Dublin</i>	1687	2240	1963.5
<i>Galway</i>	1596	2167	1881.5
<i>Mullingar</i>	1522	1663	1592.5
<i>Column Mean</i>	1556.8	2004.5	1780.6

<i>Interaction Effect</i>	<i>2007</i>	<i>2019</i>
<i>Cork</i>	-39.2	39.2
<i>Dublin</i>	-52.7	52.7
<i>Galway</i>	-61.7	61.7
<i>Mullingar</i>	A	B

- j) If there were no interaction, what values would be in this Interaction effect table?
(1 mark)
- k) Calculate the Interaction Sum of Squares (remembering that there are 10 observations per cell) and its degrees of freedom. Hence, extend the two-way ANOVA table in (f) to include the interaction term, calculating the new Error SS, Error MS and effect F-statistics. Test the significance of the interaction. What do you conclude?
(5 marks)

Question 4: [Total marks 10]

Giving up smoking can be difficult, and this difficulty can be compounded by regular alcohol consumption. To investigate the efficacy of abstaining from alcohol on smoking cessation, 20 people attempting to stop smoking were enrolled in a study. Participants were randomised to either abstaining from alcohol or not, with 10 in each group. After a month, the following 2 x 2 table of results were found

	<i>Continued smoking</i>	<i>Quit smoking</i>	<i>Total</i>
<i>Alcohol</i>	7	3	10
<i>No alcohol</i>	2	8	10
<i>Total</i>	9	11	20

- a) Compare quitting between the alcohol groups using the following measures and state what the relevant null and alternative hypotheses would be. You do not need to perform the test, simply calculate the test statistics:

- (i) Equality of proportions;
- (ii) Relative Risk;
- (iii) Odds-ratio.

(6 marks)

- b) For small studies, such as this, standard large-sample results based on the normal distribution can be unreliable. Fisher's exact test is designed to deal with this problem and calculates a randomization p-value based on the observed table and table(s) with the same marginal totals that show stronger evidence of a treatment effect. List the table(s) that are more extreme than the above table.

(4 marks)

Question 5: [Total marks 25]

Continuing the theme of smoking and alcohol, data were available from people who responded to whether they had a binge drinking session in the past week (1 = yes; 0 = no), whether they smoked (1 = yes; 0 = no) and their age (years). In the following we are interested in modelling the probability of binge drinking using logistic regression models. The output below is from a model exploring the relationship between the binary variable binge and the continuous variable age.

Fit 1: Logistic Regression Estimates Coefficients

Term	Coef	SE Coef	Z	P	Odds Ratio	95% CI	
						Lower	Upper
Constant	0.006	0.421	0.01	0.989			
age	-0.0474	0.0122	-3.87	0.000	0.95	0.93	0.98

Deviance Table

Source	DF	Adj Dev
Regression	1	15.19
age	1	15.19
Error	2716	2411.40
Total	2717	2426.59

- Write down the fitted model in both the logit and probability forms of the logistic regression model.
(3 marks)
- Interpret the estimated slope (i.e. Coef) for age and the odds ratio for age
(4 marks)
- Is there a statistically significant relationship between binge drinking and age? State the null and alternative hypotheses, compute the Wald statistic, associated p-value, and your conclusions.
(6 marks)
- Use the fitted model to predict the probability of binge drinking at age 32.
(2 marks)
- Calculate a 95% confidence interval for the slope parameter and explain how this relates to the reported interval for the Odds Ratio.
(3 marks)

Question continues on the next page...

The previous analysis (Fit 1) ignores possible differences between smokers and non-smokers. To explore this two more models were fitted as shown below.

Fit 2: Logistic Regression Estimates

Coefficients

Term	Coef	SE Coef	95% CI	Z-Value	P-Value
Constant	-0.691	0.442	(-1.556, 0.175)	-1.56	0.118
age	-0.0463	0.0127	(-0.0713, -0.0214)	-3.64	0.000
smoke 1	1.565	0.109	(1.351, 1.778)	14.36	0.000

Deviance Table

Source	DF	Seq Dev
Error	2715	2198.36
Total	2717	2426.59

Fit 3: Logistic Regression Table

Coefficients

Term	Coef	SE Coef	95% CI	Z-Value	P-Value
Constant	-1.084	0.651	(-2.361, 0.192)	-1.66	0.096
age	-0.0348	0.0189	(-0.0718, 0.0021)	-1.85	0.065
smoke 1	2.288	0.884	(0.556, 4.020)	2.59	0.010
age*smoke 1	-0.0211	0.0256	(-0.0713, 0.0291)	-0.83	0.409

Deviance Table

Source	DF	Seq Dev
Error	2714	2197.68
Total	2717	2426.59

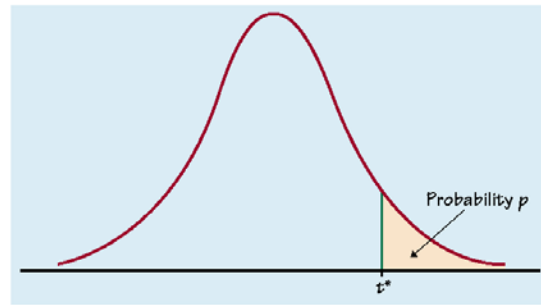
- f) Use the output (specifically the deviance values) from the three model fits to produce an analysis of deviance table for the effects of age, smoking and their interaction. What do you conclude from this table? Justify your conclusion with a formal significance test.

(5 marks)

- g) Use the interaction model estimates to calculate the odds ratio for a one-year increase in age for smokers and non-smokers.

(2 marks)

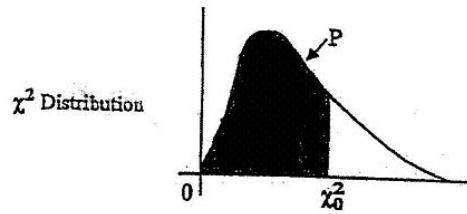
Table entry for p and C is the point t^* with probability p lying above it and probability C lying between $-t^*$ and t^* .

TABLE D t distribution critical values

df	Tail probability p											
	.25	.20	.15	.10	.05	.025	.02	.01	.005	.0025	.001	.0005
1	1.000	1.376	1.963	3.078	6.314	12.71	15.89	31.82	63.66	127.3	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60
3	0.765	0.978	1.250	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92
4	0.741	0.941	1.190	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.611	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850
21	0.686	0.859	1.063	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819
22	0.686	0.858	1.061	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792
23	0.685	0.858	1.060	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768
24	0.685	0.857	1.059	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745
25	0.684	0.856	1.058	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725
26	0.684	0.856	1.058	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707
27	0.684	0.855	1.057	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.690
28	0.683	0.855	1.056	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674
29	0.683	0.854	1.055	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646
40	0.681	0.851	1.050	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551
50	0.679	0.849	1.047	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496
60	0.679	0.848	1.045	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460
80	0.678	0.846	1.043	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416
100	0.677	0.845	1.042	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390
1000	0.675	0.842	1.037	1.282	1.646	1.962	2.056	2.330	2.581	2.813	3.098	3.300
∞	0.674	0.841	1.036	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.091	3.291
Confidence level C												
	50%	60%	70%	80%	90%	95%	96%	98%	99%	99.5%	99.8%	99.9%

F - Distribution ($\alpha = 0.05$ in the Right Tail)

df ₂	df ₁	Numerator Degrees of Freedom								
		1	2	3	4	5	6	7	8	9
Denominator Degrees of Freedom	1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54
	2	18.513	19.000	19.164	19.247	19.296	19.330	19.353	19.371	19.385
	3	10.128	9.5521	9.2766	9.1172	9.0135	8.9406	8.8867	8.8452	8.8123
	4	7.7086	9.9443	6.5914	6.3882	6.2561	6.1631	6.0942	6.0410	6.9988
	5	6.6079	5.7861	5.4095	5.1922	5.0503	4.9503	4.8759	4.8183	4.7725
	6	5.9874	5.1433	4.7571	4.5337	4.3874	4.2839	4.2067	4.1468	4.0990
	7	5.5914	4.7374	4.3468	4.1203	3.9715	3.8660	3.7870	3.7257	3.6767
	8	5.3177	4.4590	4.0662	3.8379	3.6875	3.5806	3.5005	3.4381	3.3881
	9	5.1174	4.2565	3.8625	3.6331	3.4817	3.3738	3.2927	3.2296	3.1789
	10	4.9646	4.1028	3.7083	3.4780	3.3258	3.2172	3.1355	3.0717	3.0204
	11	4.8443	3.9823	3.5874	3.3567	3.2039	3.0946	3.0123	2.9480	2.8962
	12	4.7472	3.8853	3.4903	3.2592	3.1059	2.9961	2.9134	2.8486	2.7964
	13	4.6672	3.8056	3.4105	3.1791	3.0254	2.9153	2.8321	2.7669	2.7144
	14	4.6001	3.7389	3.3439	3.1122	2.9582	2.8477	2.7642	2.6987	2.6458
	15	4.5431	3.6823	3.2874	3.0556	2.9013	2.7905	2.7066	2.6408	2.5876
	16	4.4940	3.6337	3.2389	3.0069	2.8524	2.7413	2.6572	2.5911	2.5377
	17	4.4513	3.5915	3.1968	2.9647	2.8100	2.6987	2.6143	2.5480	2.4943
	18	4.4139	3.5546	3.1599	2.9277	2.7729	2.6613	2.5767	2.5102	2.4563
	19	4.3807	3.5219	3.1274	2.8951	2.7401	2.6283	2.5435	2.4768	2.4227
	20	4.3512	3.4928	3.0984	2.8661	2.7109	2.5990	2.5140	2.4471	2.3928
	21	4.3248	3.4668	3.0725	2.8401	2.6848	2.5727	2.4876	2.4205	2.3660
	22	4.3009	3.4434	3.0491	2.8167	2.6613	2.5491	2.4638	2.3965	2.3419
	23	4.2793	3.4221	3.0280	2.7955	2.6400	2.5277	2.4422	2.3748	2.3201
	24	4.2597	3.4028	3.0088	2.7763	2.6207	2.5082	2.4226	2.3551	2.3002
	25	4.2417	3.3852	2.9912	2.7587	2.6030	2.4904	2.4047	2.3371	2.2821
	26	4.2252	3.3690	2.9752	2.7426	2.5868	2.4741	2.3883	2.3205	2.2655
	27	4.2100	3.3541	2.9604	2.7278	2.5719	2.4591	2.3732	2.3053	2.2501
	28	4.1960	3.3404	2.9467	2.7141	2.5581	2.4453	2.3593	2.2913	2.2360
	29	4.1830	3.3277	2.9340	2.7014	2.5454	2.4324	2.3463	2.2783	2.2229
	30	4.1709	3.3158	2.9223	2.6896	2.5336	2.4205	2.3343	2.2662	2.2107
	40	4.0847	3.2317	2.8387	2.6060	2.4495	2.3359	2.2490	2.1802	2.1240
	60	4.0012	3.1504	2.7581	2.5252	2.3683	2.2541	2.1665	2.0970	2.0401
	120	3.9201	3.0718	2.6802	2.4472	2.2899	2.1750	2.0868	2.0164	1.9588
	∞	3.8415	2.9957	2.6049	2.3719	2.2141	2.0986	2.0096	1.9384	1.8799



The table below gives the value x_0^2 for which $P[x^2 < x_0^2] = P$ for a given number of degrees of freedom and a given value of P .

Degrees of Freedom	Values of P									
	0.005	0.010	0.025	0.050	0.100	0.900	0.950	0.975	0.990	0.995
1	---	---	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.01	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997